

HISABDO AI

DAY 11 — AI SPENDING PATTERN EXPLANATION

Aarti — AI Explanation & Insights | Professional MVP Specification

1. Purpose

This module defines how HisabDo AI converts verified spending-pattern results from the backend into simple, natural-language insights. The AI explanation layer does not calculate financial values or independently detect patterns. It explains only the verified results supplied by the backend.

2. Aarti — Exact Task Scope

- Define the AI explanation structure for spending patterns.
- Use only verified backend spending data.
- Generate natural-language spending insights.
- Explain top spending categories.
- Explain significant spending increases/decreases.
- Generate actionable observations from verified patterns.
- Ensure AI does not invent financial numbers.
- Prepare sample AI responses for testing.

3. Correct AI Flow

HisabDo Database → .NET Spending Pattern Engine → Verified Spending Results → AI Explanation Layer → User

The .NET backend remains the source of truth for amounts, percentages, category totals, period changes, and trend classifications. The AI only converts those values into understandable language.

4. AI Explanation Structure

1. Summary — give a short overview of the user's spending pattern.
2. Top Spending — explain the highest spending categories using verified amounts/percentages.
3. Major Changes — explain significant increases or decreases compared with the previous period.
4. Pattern Observation — describe verified increasing, decreasing, stable, new, unusual, or recurring patterns.

5. Actionable Suggestion — provide a practical suggestion based only on the verified pattern.

5. Verified Backend Input

Recommended input to the AI explanation layer:

```
{
  "totalExpense": 150000,
  "topCategories": [
    {
      "category": "Food",
      "amount": 45000,
      "percentage": 30.0
    }
  ],
  "categoryChanges": [
    {
      "category": "Food",
      "currentAmount": 45000,
      "previousAmount": 35000,
      "changeAmount": 10000,
      "changePercentage": 28.57,
      "trend": "Increasing"
    }
  ],
  "recurringCandidates": []
}
```

6. Top Spending Category Explanation

For top categories, AI should explain what categories consume the largest share of spending. It should use the exact verified amount and percentage supplied by the backend.

Example:

Food is your highest spending category, accounting for 30% of your total spending in this period.

7. Significant Increase / Decrease Explanation

AI should highlight meaningful changes using the backend's trend classification and verified change percentage.

Backend Trend

AI Explanation

Action

Increasing	Spending in this category increased compared with the previous period.	Review recent expenses in this category.
Decreasing	Spending in this category decreased compared with the previous period.	Continue monitoring the category.
Stable	Spending remained broadly stable compared with the previous period.	Continue tracking the category.
New Category	This category has spending in the current period but no spending in the previous period.	Check whether this was planned or one-time spending.

8. Actionable Observations

- If a category is increasing significantly, suggest reviewing recent spending in that category.
- If a category is decreasing, acknowledge the positive change without claiming financial improvement unless the data supports it.
- If a category is stable, suggest continuing to monitor it.
- If a new category appears, suggest checking whether it is planned or one-time spending.
- If a recurring candidate is present, suggest reviewing the recurring expense and whether it is still needed.

Suggestions must remain observations/recommendations, not unsupported financial advice.

9. AI No-Invention Rules

- Never invent a transaction amount.
- Never change a backend-provided amount or percentage.
- Never calculate a different total than the backend result.
- Never invent a category or transaction.
- Never invent previous-period values.
- Never claim a pattern that is not present in the verified input.
- If a value is missing, do not guess it.
- Do not call an expense fraud or unauthorized without trusted evidence.
- Do not use outside financial data to fill missing HisabDo values.

10. Recommended AI Prompt

System:

You are the HisabDo AI spending insights assistant.

Use ONLY the verified spending data provided by the backend.

Do not invent, modify, estimate, or recalculate financial numbers.

Do not create missing transaction values.

Explain the supplied patterns in simple language.

Do not make unsupported fraud or unauthorized-transaction claims.

If information is missing, do not guess.

Return:

1. Spending Summary

2. Top Spending Categories

3. Significant Changes

4. Pattern Observations

5. Actionable Suggestions

Verified backend data:

{verified_spending_result}

11. Sample AI Responses for Testing

Sample 1 — Top Category

Your highest spending category is Food, which accounts for 30% of your total spending in this period. You may want to review recent Food expenses to understand what is driving this category.

Sample 2 — Increasing

Food spending increased by 28.57% compared with the previous period. Reviewing recent Food transactions may help identify the reason for the increase.

Sample 3 — Decreasing

Transport spending decreased by 10% compared with the previous period. This shows a lower spending pattern in Transport; continue monitoring the category.

Sample 4 — Stable

Rent spending remained stable compared with the previous period. There is no significant change in the verified spending pattern for this category.

Sample 5 — New Category

A new spending category appeared in the current period. Since there was no spending in this category during the previous period, check whether the expense was planned or one-time.

Sample 6 — Recurring

A recurring spending candidate was identified based on repeated transactions with a similar pattern. Review the expense to confirm whether it is an ongoing payment.

Sample 7 — Missing Value Safety

A spending pattern was identified, but the backend did not provide all supporting values. I will not estimate the missing financial information.

12. .NET/C# AI Explanation Preparation Code

```
using System;
using System.Collections.Generic;
using System.Linq;

public sealed record VerifiedCategorySpending(
    string Category,
    decimal Amount,
    decimal Percentage);

public sealed record VerifiedCategoryChange(
    string Category,
    decimal CurrentAmount,
    decimal PreviousAmount,
    decimal ChangeAmount,
    decimal? ChangePercentage,
    string Trend);

public sealed record VerifiedSpendingResult(
    decimal TotalExpense,
    IReadOnlyList<VerifiedCategorySpending> TopCategories,
    IReadOnlyList<VerifiedCategoryChange> CategoryChanges);

public sealed record SpendingInsight(
    string Summary,
    IReadOnlyList<string> TopCategoryInsights,
    IReadOnlyList<string> ChangeInsights,
    IReadOnlyList<string> Actions);

public sealed class SpendingAiExplanationService
{
```

```

public SpendingInsight BuildVerifiedInsight(
    VerifiedSpendingResult result)
{
    ArgumentNullException.ThrowIfNull(result);

    var summary =
        $"Total verified spending for this period is PKR {result.TotalExpense:N2}. ";

    var top = result.TopCategories
        .Take(5)
        .Select(x =>
            $"{{x.Category}} is a top spending category, " +
            $"accounting for {{x.Percentage:N2}}% of total spending " +
            $"(PKR {{x.Amount:N2}}).")
        .ToList();

    var changes = result.CategoryChanges
        .Where(x => x.Trend is "Increasing"
            or "Decreasing"
            or "New Category")
        .Select(BuildChangeInsight)
        .ToList();

    var actions = result.CategoryChanges
        .Where(x => x.Trend is "Increasing"
            or "New Category")
        .Take(5)
        .Select(x =>
            x.Trend == "New Category"
                ? $"Review {{x.Category}} spending to confirm whether it
was planned or one-time."
                : $"Review recent {{x.Category}} expenses to understand
the spending increase.")
        .ToList();

    return new SpendingInsight(
        summary,
        top,
        changes,
        actions);
}

private static string BuildChangeInsight(
    VerifiedCategoryChange change)
{

```

```

return change.Trend switch
{
    "Increasing" when change.ChangePercentage.HasValue =>
        $"{change.Category} spending increased by " +
        $"{change.ChangePercentage.Value:N2}% compared with
the previous period.",

    "Decreasing" when change.ChangePercentage.HasValue =>
        $"{change.Category} spending decreased by " +
        $"{Math.Abs(change.ChangePercentage.Value):N2}%
compared with the previous period.",

    "New Category" =>
        $"{change.Category} is a new spending category in the
current period.",

    _ =>
        $"{change.Category} has a verified spending pattern."
};
}
}

```

13. LLM Integration Boundary

The C# service above can create deterministic verified insight text without an LLM. If an LLM is later used, the .NET backend should send the verified result object to the provider and request only natural-language explanation. The LLM must not become the calculation engine.

14. Testing Scenarios

- ☐ Top category amount and percentage exactly match backend values.
- ☐ Increasing category explanation uses the verified percentage.
- ☐ Decreasing category explanation uses the verified percentage.
- ☐ Stable category is not incorrectly described as increasing/decreasing.
- ☐ New category is explained without calculating a percentage when previous amount is zero.
- ☐ Missing values are not guessed.
- ☐ AI cannot alter the backend total.
- ☐ AI cannot invent categories or transactions.
- ☐ Prompt injection inside transaction descriptions does not override system rules.

☐ User A's spending data cannot appear in User B's explanation.

15. Assumptions & Edge Cases

- Backend spending results are already validated before reaching AI.
- Top categories are already ordered by the backend.
- Trend classification is supplied by the backend.
- Previous-period amount may be zero; percentage change may therefore be unavailable.
- No transactions may exist; AI should provide a simple no-data explanation.
- Category or description may be missing.
- Recurring results should be described as candidates unless confirmed by backend business rules.

16. Aarti Deliverables

- ☐ Approved spending explanation structure.
- ☐ Verified backend input contract.
- ☐ Prompt and no-invention rules.
- ☐ Natural-language insight templates.
- ☐ Top-category explanation templates.
- ☐ Increase/decrease explanation templates.
- ☐ Actionable observation templates.
- ☐ Sample AI responses.
- ☐ AI response test cases.

17. Team Lead Acceptance Checklist

- ☐ AI explanation structure is approved.
- ☐ Only verified backend data is used.
- ☐ Top categories are explained correctly.
- ☐ Significant increases/decreases are explained correctly.
- ☐ Actionable observations are based on verified patterns.
- ☐ AI cannot invent or modify financial numbers.
- ☐ Sample responses pass testing.
- ☐ Missing values and edge cases are handled safely.

18. Final Definition

Aarti's Day 11 AI Explanation module converts verified spending-pattern results into natural-language insights for HisabDo users. It explains top categories, significant increases/decreases, recurring or notable patterns, and practical observations. All financial facts originate from the .NET backend; AI is used only as the explanation layer and must never invent or modify financial numbers.